

HAT-P-54b

R-1843

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_160322 · created 2026-08-03T07:11:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457469.65675 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.041
Transit depth (Rp/Rs)^2	1.15 +/- 0.89 [%]
Orbital Inclination (inc)	86.67 +/- 2.46
Airmass coefficient 1 (a1)	1.046 +/- 0.025
Airmass coefficient 2 (a2)	-0.042 +/- 0.049
Scatter in the residuals of the lightcurve fit is	2.34 %
Best Comparison Star	#2 - [500, 24]
Optimal Aperture	2.33
Optimal Annulus	10.33
Transit Duration (day)	0.058 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.107 $\pm$ 0.041 vs 0.1572 (deviation 0.9 σ)
Duration fit vs published	0.058 d vs 0.075 d (deviation 0.78 σ)
Mid-time O-C	-4.3 min
Depth SNR	1.9
Residual scatter	2.34 %

Light curve

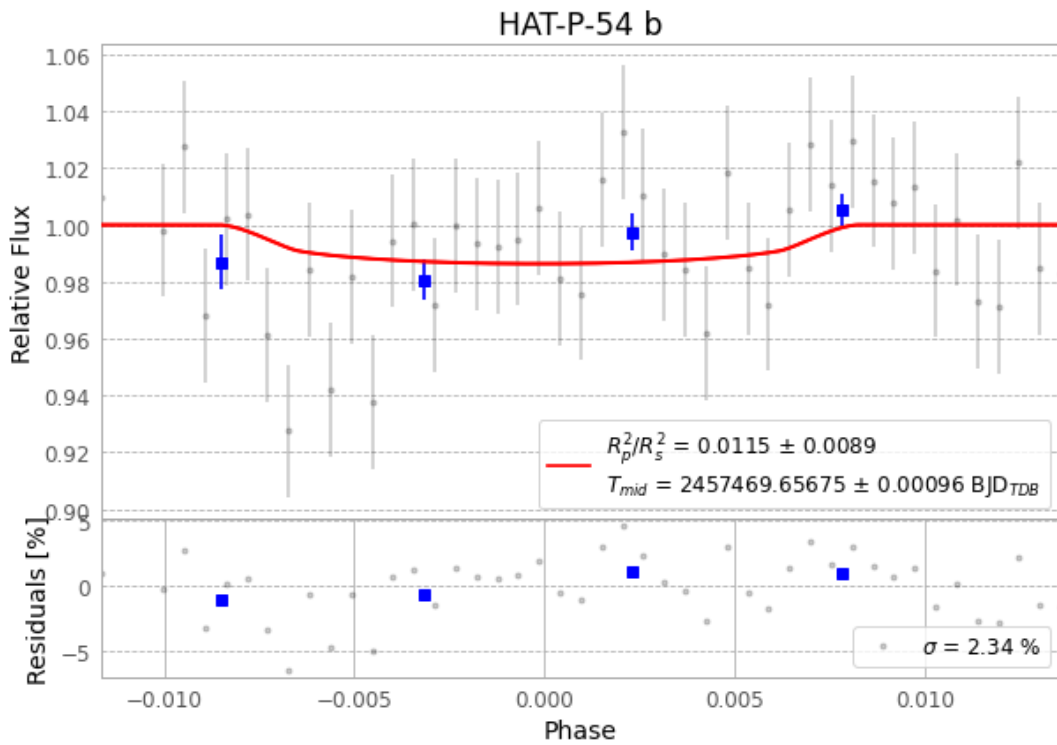


Figure 1 — FinalLightCurve_HAT-P-54 b_2016-03-21.png (SHA-256 3ee863cc51654808...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [196, 252], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e69bc9184ead1971...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

46 FITS frames (30 MB), HATP-54160322023915.FITS → HATP-54160322050012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-160322.log (51ca25273ecd...), FinalLightCurve_HAT-P-54 b_2016-03-21.png (3ee863cc5165...), FinalLightCurve_HAT-P-54 b_2016-03-21.csv (bd8bcab59eae...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 07:11Z.